

Lecture 1: Finding Solutions

Introduction to Data Analysis Techniques

1 Data Analysis in Science

Definition 1.1 (Analytical Method and Analytical Solution). [def:analytical-method] When we use *analytical methods* to solve a problem, we are manipulating the mathematical expressions until unknowns are being isolated into a standard mathematical form.

An *analytical solution* is a solution for a problem such that the unknowns are expressed in a closed form.

Example 1.1 (Analytical Solution of a Quadratic Equation). [eg:quadratic-equation] Consider the quadratic equation

$$ax^2 + bx + c = 0, \quad (1)$$

where a , b , and c are constants. The analytic solution for x can be found using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (2)$$

Definition 1.2 (Numerical Method and Numerical Solution). [def:numerical-method] *Numerical methods* are methods to obtain numerical solutions.

A *numerical solution* is a solution for a problem such that the unknowns are expressed in an numerical form.

Example 1.2 (Numerical Solution of a Trigonometric Equation). [eg:trig-equation] Consider the trigonometric equation

$$x = \beta \cos x \quad (3)$$

where β is a constant.

It is very difficult to find an analytical solution for this equation using algebraic manipulations. However, we can use numerical methods to find an approximate solution for specific β values.

The numerical solutions can be

$$\beta = 1, \quad x \approx 0.73909, \quad \beta = 0.5, \quad x \approx 0.4508$$

Remark 1.1 (Advantages and Disadvantages of Numerical Methods). [rem:numerical-methods]

1. For numerical methods, we must have all parameters of the problem known.
2. If parameters change, solution must be re-done.
3. Solutions may be not exact.
4. Numerical solutions often considered inferior to analytic solutions, if exist. *ie, we should use analytical methods whenever possible or too difficult.*

2 Using Taylor Series to Approximate Functions

Definition 2.1 (Taylor Series). [def:taylor-series] For a function $f(x)$ that is infinitely differentiable at a point a , then:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Remark 2.1 (Taylor Series Approximation). [rem:taylor-series-approx] It is impossible to compute the infinite sum in practice. When we find the analytical solution of a problem, we can use the first few terms of the Taylor Series to approximate the function.

The more terms we use, the better the approximation.

Example 2.1 (Electric Field of a Dipole for $x \gg a$). [eg:dipole-field] Consider the electric field on the axis of a dipole with charges $+q$ at $x = a$ and $-q$ at $x = -a$:

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(x-a)^2} - \frac{1}{(x+a)^2} \right]$$

Take the Taylor series expansion of E for $x \gg a$:

$$E = \frac{q}{4\pi\epsilon_0 x^2} \left[4\frac{a}{x} + 8\left(\frac{a}{x}\right)^3 + 12\left(\frac{a}{x}\right)^5 + \dots \right]$$

Keeping the first two terms gives an approximation for $x \gg a$:

$$E \approx \frac{q}{4\pi\epsilon_0 x^2} \left[4\frac{a}{x} + 8\left(\frac{a}{x}\right)^3 \right]$$

Which means, for $x \gg a$, $E \propto x^{-3}$.

3 Fixed Point Iteration Method

Definition 3.1 (Fixed Point Iteration Method). [def:fixed-point-iteration-method] The idea for *fixed point iteration method* is to first find a function $g(x)$ such that root r is a fixed point of $g(x)$, i.e., $g(r) = r$